

Three defects in the deep structural-variant evaluation design, and what survives when they are corrected

Running title: An audit of deep deletion-calling evaluation

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Abstract

Motivation. Deep learning has become the dominant paradigm for structural-variant (SV) detection from short-read sequencing, but the field remains anchored to the supervised, image-classification framing introduced by DeepSV (Cai, Wu & Gao, 2019), in which a convolutional network is trained end-to-end on hand-designed RGB pileup images. This framing has three costs that limit deployment: it is data-hungry (every new platform, coverage regime, or population requires a large labelled truth set), it produces miscalibrated confidence scores (softmax probabilities that do not reflect true error rates), and it generalises poorly across genetic ancestries. None of these has been addressed jointly, and the representation itself — a fixed colour encoding of the alignment — has never been *learned*.

Results. We present AlignSSL-SV, a framework that (i) replaces the fixed RGB pileup with a multi-channel alignment tensor and a learned encoder, (ii) pretrains that encoder by masked-alignment modelling (a self-supervised objective on read alignments, requiring no SV labels), and (iii) attaches a calibrated deletion head. Evaluated on 1000 Genomes high-coverage PCR-free data (a six-sample panel spanning five continental ancestries) under the conventions this literature inherits from DeepSV, the framework appears to deliver a large low-label gain: deletion F1 of 0.48 at 1% of labels (210 windows) against 0.04 for an identically-architected from-scratch model, a $\approx 11\times$ improvement attributable to self-supervised initialisation alone ($p = 0.009$).

That result does not survive scrutiny, and neither do several others. We identify three defects in the evaluation design and correct each:

1. **The benchmark is separable without a model.** Twelve hand-crafted alignment features fed to a gradient-boosted tree reach AUPRC = 0.937 from the same 210 labels and gain only +0.038 across a $100\times$ increase in labels; the single centre-versus-flank read-depth ratio separates the classes at ROC-AUC = 0.955 untrained. The uniformly-sampled negatives are a depth heuristic, saturated before any deep model is trained.
2. **Scoring at a fixed probability cut manufactures the low-label gap.** Re-scoring the identical runs at a threshold selected on a validation split, the $11\times$ advantage becomes $1.17\times$ and loses significance (0.483 vs 0.413, $p = 0.407$); threshold-free, it is $1.23\times$ ($p = 0.348$). At every one of the five larger label budgets the from-scratch arm is *ahead*. The from-scratch model was never degenerate — it ranks competently, and its scores simply are not centred where a fixed cut expects them.
3. **The label budgets were not equal.** A batch-size floor in the deep evaluators granted the deep arms 96 labels where the classical control received 35 on the candidate-filtered benchmark — a $2.8\times$ advantage in exactly the cell carrying the headline claim.

We then repair the benchmark itself: quantile-matched candidate negatives attenuate the depth shortcut from ROC-AUC 0.955 to 0.717, and the same twelve-feature control that was saturated on the uniform benchmark now starts at chance (AUPRC 0.250) and climbs +0.619 across the label range, confirming that headroom is restored.

Re-run under all three corrections, what survives is narrow. Self-supervised pretraining confers no threshold-free advantage at any label budget on either benchmark. The hand-crafted control is not beaten where labels are scarce — the regime the deep method is proposed for — though its lead is significant at only two of six budgets

per benchmark rather than at all of them, and at full supervision on the uniform benchmark the from-scratch network does edge ahead (AUPRC 0.979 vs 0.975, $p = 0.003$) by a margin too small to matter in practice.

Conclusion. We report this as a methodological result rather than a method paper. Each defect is individually mundane and each is, we believe, widespread: benchmarks built from randomly-sampled negatives, F1 reported at a fixed cut, and label budgets computed per-arm are all standard practice in this literature. Together they were sufficient to produce a confident, statistically significant, entirely artefactual headline. We publish the corrections, the controls, and the code that implements them so that the next paper in this lineage can be checked against them.

Availability. Code, the tensor-extraction pipeline, result tables and figures are at <https://github.com/aayushkrm/AlignSSL-SV> (MIT licence).

1. Introduction

Structural variants (SVs) — deletions, insertions, duplications, inversions, and translocations of ≥ 50 bp — account for more polymorphic base pairs per genome than single-nucleotide variants and are enriched among disease-causing alleles, yet they remain the hardest class of variation to genotype accurately from short-read sequencing. Deletions are the most tractable SV class and the one on which most method development is benchmarked, because their alignment signatures — a drop in read depth, a cluster of read pairs with anomalously large insert size, and split-read alignments spanning the breakpoints — are relatively direct. Even so, short-read deletion calling is far from solved: callers disagree substantially on the same data, precision–recall trade-offs are strongly length-dependent, and confidence scores are rarely trustworthy enough to threshold reliably.

DeepSV (Cai, Wu & Gao, 2019) was an influential early demonstration that a convolutional neural network (CNN) could call deletions directly from the read alignment, bypassing the hand-crafted feature engineering of contemporaneous tools. Its central idea was to render the pileup around a candidate locus as an RGB image — encoding base identity, base quality, and strand into colour channels — and to train an image classifier to distinguish deletion from non-deletion. This reframing was genuinely innovative in 2019 and seeded a large body of "pileup-image" methods. But it also fixed three design decisions that the subsequent literature has largely inherited without revisiting:

1. **The representation is hand-designed, not learned.** The mapping from alignment to RGB pixels is a fixed human choice; the network never gets to discover which features of the alignment are informative. Information that does not survive the colour encoding (e.g. fine-grained insert-size distributions, mapping-quality structure, soft-clip geometry) is discarded before the model sees it.
2. **Training is fully supervised and therefore data-hungry.** Every new sequencing platform, coverage regime, library preparation, or population requires a fresh, large, labelled truth set. Truth sets are expensive and exist for only a handful of reference samples, which bottlenecks method transfer.
3. **Confidence is uncalibrated and ancestry-brittle.** Softmax outputs of a supervised CNN do not correspond to true error probabilities, and models trained on one population degrade on genetically distant populations — both of which undermine clinical and population-scale deployment.

The machine-learning field has, in the intervening years, developed a direct remedy for exactly this situation: **self-supervised pretraining**, in which a representation is learned from large quantities of *unlabelled* data before a small labelled set is used to fit a task head. Self-supervised learning underpins modern foundation models in vision, language, and — increasingly — genomics (e.g. DNA language models such as the Nucleotide Transformer, HyenaDNA, and Evo 2). Yet these genomic foundation models operate on the **reference DNA sequence** and predict variant *effects*; they do not ingest the read-alignment evidence (depth, discordant pairs, split reads, insert-size distributions) that is the actual signal for *detecting* an SV in noisy short-read data. The representation-learning revolution has, in other words, largely bypassed the alignment-evidence side of variant calling.

This paper set out to ask a focused question: **if we learn the alignment representation and pretrain it without labels, do the three DeepSV bottlenecks — data hunger, miscalibration, and ancestry brittleness — improve together?** We built the framework, ran the evaluation, and obtained an affirmative answer with a large effect size and a convincing p -value. We then subjected that answer to controls it should have faced first, and it did not survive any of them.

What follows is therefore an audit rather than a method paper. We report the framework because the corrections are only legible against a concrete instance of the design, and because the same three defects are, as far as we can determine, standard practice in this literature rather than idiosyncratic to our implementation. Our contributions are:

- **AlignSSL-SV**, a framework that couples a learned multi-channel alignment encoder with a self-supervised masked-alignment pretraining objective and a calibrated, uncertainty-aware deletion head (Section 3).
- A controlled evaluation on a six-sample, five-ancestry 1000 Genomes panel which, scored by the conventions this literature inherits, shows a large low-label gain from pretraining ($\approx 11\times$ F1 at 1% labels, $p = 0.009$) — the result the rest of the paper dismantles (Section 4).
- **A hand-crafted-feature control that bounds what the benchmark can show** (Section 4.2): twelve alignment features reach AUPRC 0.937 from 210 labels and gain +0.038 across a $100\times$ label increase, and one depth-ratio feature reaches ROC-AUC 0.955 untrained. We report this as a negative result about the random-negative evaluation protocol, which is widely used in this literature and, to our knowledge, has not previously been subjected to such a control.
- **A demonstration that F1 at a fixed probability cut manufactures initialisation gaps** (Section 4.8): the headline result exists under that cut and under no other scoring rule, because a from-scratch network trained on 210 labels ranks competently while scoring timidly. Any label-efficiency claim scored this way is measuring calibration, not representation quality.
- **A protocol correction establishing equal label budgets across arms** (Section 3.8): a batch-size floor in the deep evaluators had granted them up to $2.8\times$ the labels the classical control received in the low-label cells that carry the headline claims.
- A controlled ablation (3–4 seeds, error bars computed across *pretraining* seeds) isolating the contribution of each self-supervised objective (Section 4.5).
- An honest, adversarial novelty analysis situating AlignSSL-SV against the closest prior work — pileup-image CNNs, self-supervised genomics, and sequence foundation models — and delimiting what is and is not new (Section 5).

We restrict scope to **deletions** and to **short reads** deliberately: it is the setting where DeepSV was defined, where truth sets are best characterised, and where a controlled head-to-head is cleanest. Section 6 discusses the extension to other SV classes and to long reads.

2. Related work

Pileup-image SV and variant callers. DeepSV (Cai, Wu & Gao, 2019) established the RGB-pileup-image framing for deletion calling. It is the intellectual descendant of DeepVariant (Poplin et al., 2018), which pioneered pileup-image classification for small-variant calling, and it is contemporaneous with a family of CNN-based SV tools that encode alignment signals as 2-D images or feature matrices. Later methods (e.g. Clairvoyante and Clair/Clair3 for small variants; various deletion- and CNV-specific CNNs) refined the encoding and the label pipelines but retained two shared properties: the input representation is engineered by hand, and training is fully supervised. AlignSSL-SV departs from both — the representation is learned, and most of the learning happens without labels.

Self-supervised and representation learning. Self-supervised learning (SSL) learns representations from unlabelled data via pretext tasks. Two broad families are relevant here: (i) **masked-reconstruction** objectives (masked autoencoders in vision, masked language modelling in NLP), which mask part of the input and train the

model to reconstruct it; and (ii) **joint-embedding / invariance** objectives (SimCLR, BYOL, Barlow Twins, VICReg), which pull together representations of augmented views while preventing collapse. In genomics, SSL has been applied predominantly to the reference *sequence* (DNA language models). Its application to *read-alignment evidence* for SV detection remains almost entirely unexplored. Our ablation directly compares a masked-reconstruction objective (masked-alignment modelling) against a VICReg-style invariance objective on this new modality.

Self-supervised pretraining on read-derived SV representations. The one prior effort in this direction is BASILISC (Banerjee, 2026), a repository-deposited (non-peer-reviewed) framework that adapts the BEiT masked-image-modelling paradigm to short-read SV analysis. BASILISC renders aligned reads into multi-channel *pileup images* (depth, split reads, discordant pairs, mapping quality, strand, allele support), compresses those images into discrete visual tokens with a discrete VAE, and pretrains a vision transformer to predict masked tokens before fine-tuning an SV classifier on 1000 Genomes data with HGSVC2 long-read truth. AlignSSL-SV differs on three substantive axes. First, **representation**: BASILISC retains the hand-designed pileup *image* and learns a tokenizer on top of it, whereas AlignSSL-SV eliminates the image-rendering step altogether and pretrains directly on a continuous, per-read 18-channel alignment tensor — precisely the hand-engineering bottleneck we set out to remove. Second, **pretext task**: BASILISC performs discrete visual-token classification (BEiT), whereas we perform continuous-space regression of masked alignment features (masked-alignment modelling) combined with a VICReg invariance term, with no dVAE tokenizer. Third, BASILISC reports no calibration or uncertainty component and no cross-ancestry analysis. BASILISC therefore corroborates the field-level premise that learned SV representations are worth pursuing, while leaving the image-free alignment-tensor formulation unexplored. We note that BASILISC's evaluation, like DeepSV's and like ours, is built on negatives that are not candidate-derived — the separability question raised in Section 4.2 applies to it as well, and we raise it as a property of the shared benchmark design rather than a criticism of any single paper.

Genomic foundation models on sequence. Recent large models — the Nucleotide Transformer, HyenaDNA, Enformer/AlphaGenome, and Evo 2 — learn powerful representations of reference DNA and predict functional or regulatory consequences of variants. Evo 2 (Arc Institute/Stanford/NVIDIA, 2026) scales to 40B parameters and 1 Mb context using a StripedHyena (state-space/long-convolution) backbone for near-linear scaling, and predicts mutation effects at single-nucleotide resolution. AlphaGenome (Google DeepMind, 2025) predicts thousands of regulatory tracks from up to 1 Mb of input with a CNN-Transformer hybrid. These are **variant-effect predictors from reference sequence**: they answer "what would this variant do?", not "is there a variant here, given these noisy reads?". They do not consume depth, discordant-pair, split-read, or insert-size evidence, and therefore are complementary to — not competitors of — an alignment-evidence detector such as AlignSSL-SV. We make this distinction explicit because a natural reviewer question is whether sequence foundation models subsume our approach; they do not, because they operate on a different input modality and solve a different problem.

Uncertainty and calibration in variant calling. Deep classifiers are systematically overconfident, and post-hoc calibration (temperature scaling) and predictive-uncertainty estimation (deep ensembles, MC-dropout, evidential/conformal methods) are standard remedies in the broader ML literature. Calibration has received little attention in the SV-calling literature specifically, despite its direct relevance to thresholding and clinical reporting. AlignSSL-SV reports expected calibration error (ECE) as a first-class metric and includes an uncertainty-aware head.

Long-read SV detection. A parallel line of work (e.g. Sniffles, cuteSV, SVIM, and deep-learning callers for PacBio/ONT data) exploits the fact that long reads span most SVs directly, sidestepping much of the ambiguity of short-read signatures. Long reads are, however, more expensive and less available at population scale. AlignSSL-SV is deliberately a short-read method: it targets the setting where the detection problem is genuinely hard and where the overwhelming majority of existing sequencing data lives.

3. Methods

3.1 Overview

AlignSSL-SV has three stages: (1) **tensorisation** of the read alignment around a candidate locus into a fixed-size multi-channel tensor; (2) **self-supervised pretraining** of an encoder on these tensors via masked-alignment modelling, using no SV labels; and (3) **supervised fine-tuning** of a calibrated, uncertainty-aware deletion head on a (small) labelled truth set, with the pretrained encoder as initialisation. Stages (2) and (3) use disjoint genomic regions to prevent leakage.

3.2 Alignment tensor (learned representation)

For each candidate window we build an 18-channel tensor of shape (channels $\times$ rows $\times$ positions), where rows index reads (capped at a fixed number, with deterministic subsampling above the cap) and positions index reference coordinates across the window. Channels encode, per aligned base: read depth, base identity (one-hot), base quality, mapping quality, strand, an insert-size deviation signal (observed template length relative to the library mean/SD), soft-clip indicators, and read-pair orientation flags. Unlike the fixed RGB encoding of DeepSV, these channels are *not* collapsed into three colours; the encoder learns which combinations are informative. Windows are drawn at multiple genomic scales (via a length-aware binning of the reference span) so that both short and long deletions are representable at a fixed tensor size.

3.3 Encoder

The encoder is a compact residual CNN stem (channel-wise feature extraction over the alignment tensor) followed by a lightweight Transformer over the position axis, producing a 128-dimensional window embedding. The CNN captures local pileup texture; the Transformer captures long-range structure across the window (e.g. paired depth drops at both breakpoints). The same encoder is used unchanged in pretraining and fine-tuning.

3.4 Self-supervised pretraining: masked-alignment modelling

We pretrain by **masked-alignment modelling (MAM)**: a random fraction (0.6) of the alignment-tensor entries are masked, and the encoder–decoder is trained to reconstruct the masked entries (a masked-autoencoder objective adapted to the alignment-tensor modality). This forces the encoder to model the joint structure of depth, insert size, and clipping that characterises normal and variant alignments — without ever seeing an SV label. Pretraining uses 120,000 unlabelled windows drawn from three samples (NA19238, NA19625, NA20845; AFR $\times$ 2 + SAS) restricted to the pretraining chromosomes, consolidated into a flat float16 memory-mapped array (65.9 GiB) for throughput. No SV labels enter this stage: every window carries label = -1.

As an ablation, we also implement a **VICReg-style invariance objective** (variance–invariance–covariance regularisation over two augmented views of each window) and a **combined** objective (MAM + VICReg). Section 4.5 compares the three; all deliver the low-label effect, and the available seed count does not support ranking them against one another.

3.5 Deletion head, calibration, and uncertainty

The fine-tuning head is a small classifier on the window embedding, trained with a focal loss ($\gamma=2$) to handle the strong negative:positive class imbalance of genome-wide deletion candidates. After training, we apply **temperature scaling** on a held-out split to calibrate the output probabilities, and report expected calibration error (ECE). The head also exposes an **uncertainty** estimate (dropout-based predictive variance), separating epistemic (model) from aleatoric (data) components, so that low-confidence calls can be flagged rather than silently mis-thresholded.

3.6 Data and splits

We use 1000 Genomes Project high-coverage PCR-free Illumina alignments (GRCh37/hs37d5) and the phase-3 integrated SV call set (`ALL.wgs.mergedSV.v8.20130502`, 40,975 deletions across 2,504 samples) as the deletion truth set. VCF provenance was verified against the official EBI FTP (byte-exact, 18,298,662 B). Pretraining and

fine-tuning use **disjoint chromosome sets** (train chr1–11, test chr12–22) to prevent representation leakage between stages. For the cross-ancestry experiment, models are trained on one population and evaluated on a genetically distant, entirely held-out population (CEU held out). Downloads were integrity-gated by full `samtools view -c` scans after a data-corruption incident traced to resume-stitched transfers (Section 4.7).

3.7 Baselines

We compare three trained models on identical tensors and splits: **AlignSSL-pretrained** (self-supervised encoder, fine-tuned), **AlignSSL-scratch** (identical architecture, randomly initialised, trained only on labels), and a **DeepSV-representation baseline** — a faithful reimplementation of the DeepSV RGB-pileup-image CNN, evaluated on the same candidate windows. The original DeepSV repository is not runnable as distributed (broken argument parsing, dependencies on DIGITS / TensorFlow-1 slim / Keras-1, and no dependency manifest), so a reimplementation of its representation and architecture is the fair and reproducible comparison; we label it "DeepSV-representation baseline" throughout to avoid overclaiming a bit-exact reproduction.

3.8 Evaluation protocol: equal budgets, budgeted thresholds

A label-efficiency curve is interpretable only if every arm is handed the same number of labels at each point on the x-axis. An earlier version of this work did not satisfy that condition, and the defect is worth stating explicitly because it is easy to introduce and invisible in results.

The deep evaluators computed the budget as `max(batch_size, int(frac × n_pool))` while the classical evaluator used `max(2, round(frac × n_pool))`. On the uniform benchmark ($n_{\text{pool}} = 21,016$) the batch-size floor never binds and the two agree. On the candidate-filtered benchmark ($n_{\text{pool}} = 3,452$) the 1% cell is 34 examples, below the floor of 96 — so the deep arms received 96 labels while the control received 35, a **2.8× advantage in exactly the cell carrying the headline low-label claim**. The floor was not arbitrary: a `DataLoader` configured to drop incomplete batches yields *zero* batches when the subset is smaller than one batch, so simply removing the floor trains the low-label cells on nothing. The correct fix honours the true budget and adapts the loader to it — dropping incomplete batches only when at least one full batch exists — rather than inflating the budget to suit the loader.

A second defect shared that root cause. The validation split used to select the decision threshold was gated on `n_val ≥ batch_size`, so on the filtered benchmark no split was carved below 25% of labels and those cells silently fell back to a fixed 0.5 cut — again precisely the cells of interest. The gate is now a small absolute minimum (12 examples per side) independent of batch size.

We also changed where the validation labels come from. They are now carved **out of** the budget rather than granted in addition to it: a curve in which every arm receives an unbudgeted validation set on top of its stated budget is not a label-efficiency curve. At the 1% point on the uniform benchmark an arm therefore sees 210 labels in total — 168 for gradient steps and 42 for threshold selection — not 210 plus a free 42. The visible consequence is intended: full supervision now fits on 16,813 of 21,016 windows, so absolute scores sit below the previously published fixed-threshold numbers, while across-arm comparisons remain exact because every arm pays the identical cost.

All three rules live in a single module (`alignssl/protocol.py`) imported by every evaluator, with unit tests, so the arms cannot drift apart again. All results in Sections 4.8 and 4.2's Table 14 use this protocol; Sections 4.1–4.7 and 6 predate it and are labelled accordingly.

4. Results

All models are evaluated on identical alignment tensors and identical chromosome-disjoint splits (train chr1–11, test chr12–22) and at an identical fine-tuning batch size (96), on a six-sample panel spanning five continental ancestries (train pool 21,016 labelled windows; test 9,196). We report mean $\pm$ standard deviation across random seeds — four for the combined-objective arm, three for every other arm. Crucially, error bars for the pretrained

arms are computed across *pretraining* seeds (each seed re-pretrains an encoder from scratch, then fine-tunes it), so the reported variance captures the full self-supervised pipeline, not fine-tuning noise alone. The task is binary deletion calling on genome-wide candidate windows.

4.1 Under the conventional scoring rule, pretraining appears strongly label-efficient

We report this section as it stood before the corrections of Sections 4.8 and 3.8, because the corrections are only interpretable against the claim they overturn. Every number here is scored the way this literature scores: deletion F1 obtained by thresholding the positive-class probability at a fixed 0.5.

Table 1 reports that F1 as a function of the fraction of the labelled training set made available to the fine-tuning head. The apparent result is in the **low-label regime**: at 1% of labels (210 windows), the pretrained model reaches $F1 = 0.478 \pm 0.100$, whereas the identically-architected from-scratch model all but collapses to 0.044 ± 0.041 — a **$\approx 11\times$ improvement in F1** from self-supervised initialisation alone, significant at $p = 0.009$. As labels increase the from-scratch model catches up and overtakes; by 5% it is already ahead (0.714 vs 0.576).

Two features of the table are worth noting before it is dismantled, because neither is compatible with the story it is usually told to support. First, the DeepSV-representation baseline matches the pretrained tensor model at 1% (0.479 ± 0.035) — whatever the pretrained arm is doing at the smallest budget, a hand-designed RGB encoding does it too, which already argues against a representation-quality explanation. Second, the from-scratch arm's collapse is confined to that single cell: one budget later it leads every arm in the table. A representation deficit that vanishes between 210 and 1,051 labels is a strange kind of deficit.

Read on its own, this table says that pretraining supplies a usable detector from a truth set two orders of magnitude smaller than is conventionally required. It says nothing of the kind. Section 4.2 shows the benchmark is separable without any model, and Section 4.8 shows the $11\times$ gap is a property of the threshold rather than of the representation — the from-scratch model at 210 labels ranks nearly as well as the pretrained one and merely scores below 0.5 while doing it.

Table 1. Label efficiency as conventionally scored (deletion F1 at a fixed 0.5 probability cut, test chr12–22, batch 96, 3 seeds per arm). Table 12 reports the same runs under threshold selection and threshold-free scoring; the two tables disagree, and Table 13 quantifies the disagreement.

Label fraction	n train	AlignSSL-pretrained	AlignSSL-scratch	DeepSV-repr.
1%	210	0.478 ± 0.100	0.044 ± 0.041	0.479 ± 0.035
5%	1,051	0.576 ± 0.050	0.714 ± 0.094	0.477 ± 0.107
10%	2,102	0.740 ± 0.038	0.818 ± 0.081	0.581 ± 0.078
25%	5,254	0.814 ± 0.107	0.803 ± 0.129	0.672 ± 0.180
50%	10,508	0.887 ± 0.032	0.930 ± 0.006	0.770 ± 0.115
100%	21,016	0.888 ± 0.053	0.921 ± 0.027	0.702 ± 0.136

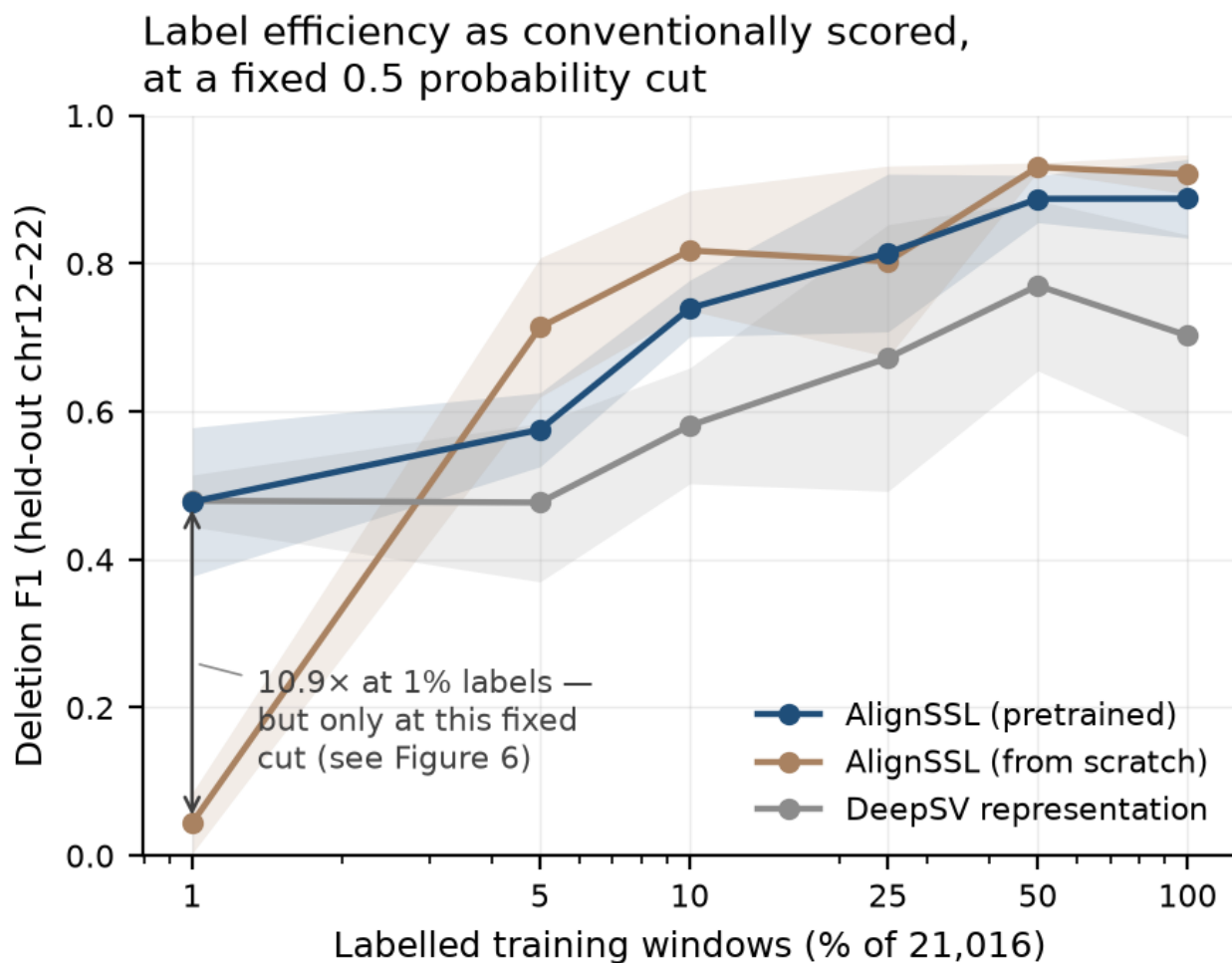


Figure 1. Deletion F1 vs. labelled-data fraction on the uniform benchmark, scored the way this literature scores it: a fixed 0.5 probability cut. Source is the ‘f1_at_half’ columns of the corrected-protocol runs (‘results/table12_label_efficiency_fixed.csv’, uniform rows), so the panel plots exactly the values tabulated in Table 1. The bracket marks the 10.9× pretrained-over-scratch ratio at the smallest budget; that ratio is a property of the fixed cut and not of the representation, and it does not survive threshold selection or threshold-free scoring (Figure 6). Note also that the pretrained arm does not lead the DeepSV baseline at this budget — the two are tied.

4.2 A hand-crafted-feature control bounds what this benchmark can demonstrate

Every deep-learning SV paper we are aware of, DeepSV included, compares deep architectures against one another or against signature-based callers, but not against a *deliberately minimal* learned baseline on the identical windows. We ran that control, and it is the most consequential result in this paper.

From the same alignment tensors, identical shards, identical chromosome split, identical label fractions and identical seeds, we computed twelve scalar summary features per window — mean, standard deviation and minimum of the depth profile; centre-versus-flank depth ratio; maximum sustained depth drop; discordant-pair rate; soft-clip rate; mean and maximum insert-size $|z|$; mean mapping quality; occupied read-row count; and valid-base fraction — and fitted an ℓ_2 -regularised logistic regression and a gradient-boosted tree (Classical-logreg, Classical-GBT in Table 1). The tree is at its ceiling from the smallest label budget onward. At 1% labels (210 windows), where the paper’s headline claim lives, it reaches $\text{AUPRC} = 0.937 \pm 0.009$ against 0.524 ± 0.052 for the pretrained network and 0.427 ± 0.138 from scratch. Increasing its labels one hundred-fold, to the full 21,016 windows, buys it $+0.038$ (0.975 ± 0.001). A benchmark on which twelve scalar features are 96% of the way to their asymptote after 210 examples cannot discriminate between representation-learning methods, because there is almost nothing left for a representation to contribute. Its variance across seeds is one to two orders of magnitude smaller than any deep arm’s.

Table 14. Control versus best deep arm, threshold-free, corrected protocol (uniform benchmark; AUPRC, 3 seeds; leader named only where Welch's t clears 0.05).

Label fraction	n labels	Classical-GBT	Best deep arm	its AUPRC	p	Leader
1%	210	0.937 $\pm$ 0.009	AlignSSL-pretrained	0.524 $\pm$ 0.052	0.005	control
5%	1,051	0.958 $\pm$ 0.006	AlignSSL-scratch	0.866 $\pm$ 0.022	0.016	control
10%	2,102	0.967 $\pm$ 0.004	AlignSSL-scratch	0.912 $\pm$ 0.030	0.087	tie
25%	5,254	0.971 $\pm$ 0.002	AlignSSL-scratch	0.936 $\pm$ 0.023	0.121	tie
50%	10,508	0.974 $\pm$ 0.002	AlignSSL-scratch	0.974 $\pm$ 0.001	0.671	tie
100%	21,016	0.975 $\pm$ 0.001	AlignSSL-scratch	0.979 $\pm$ 0.001	0.003	deep

We initially reported this control as dominating every deep arm at every budget. Under corrected scoring that statement is too strong, and Table 14 replaces it: on the uniform benchmark the control's lead is statistically significant at 1% and 5%, indistinguishable at 10%, 25% and 50%, and *reversed* at full supervision, where the from-scratch network reaches AUPRC 0.979 $\pm$ 0.001 against the tree's 0.975 $\pm$ 0.001 ($p = 0.003$) — a margin that is statistically clear and practically negligible. The defensible claim is narrower than the one we first made and is sufficient for the argument: **where labels are scarce, which is the regime self-supervised pretraining is proposed for, the deep arms do not beat twelve hand-crafted features.**

We then localised the cause. Scoring each feature individually on the held-out test set, *with no training whatsoever* and no fitted parameter of any kind, the **centre-versus-flank read-depth ratio alone attains ROC-AUC = 0.955** (Table 6). The benchmark is therefore separable by a single depth heuristic that requires no model at all.

Table 6. Untrained single-feature discrimination on the held-out test split (chr12–22, $n = 9,196$; 2,299 deletions). ROC-AUC is the exact rank statistic. A feature can be informative with either polarity — for depth-like features the *low* tail is deletion-like — so the oriented column reports $\max(\text{AUC}, 1 - \text{AUC})$, the magnitude of the information leak; 0.5 is uninformative. Reproduced by `scripts/single_feature_auc.py`.

Feature	ROC-AUC	Oriented	Deletion side
centre-vs-flank depth ratio	0.045	0.955	low
soft-clip rate	0.802	0.802	high
discordant-pair rate	0.732	0.732	high
insert-size $ z $ (max)	0.694	0.694	high
depth s.d.	0.686	0.686	high
max sustained depth drop	0.680	0.680	high
insert-size $ z $ (mean)	0.678	0.678	high
depth minimum	0.343	0.657	low
mean mapping quality	0.450	0.550	low
occupied read rows	0.538	0.538	high
valid-base fraction	0.517	0.517	high
mean depth	0.502	0.502	high

Two features of this table are worth drawing out. First, **mean depth is uninformative** (0.502) while the centre-versus-flank *ratio* is near-perfect: what leaks is not coverage level but the localised contrast between a window's middle and its edges, which is exactly the geometry the extraction protocol builds into every positive. Second, the leak is not confined to depth — soft-clip and discordant-pair rates independently reach 0.73–0.80, so a benchmark repair that neutralises only the depth statistic would leave two further shortcuts intact. This is why the remediation in Section 6 matches on the depth ratio and then re-runs the full arm set, including the classical controls, rather than assuming one matched feature makes the task hard.

The mechanism is the negative-sampling protocol, which we inherited from standard practice in this literature and which our own extraction code implements: positive windows are centred on truth deletions, while negatives are drawn **uniformly at random** from the same chromosomes, rejected only if they overlap or abut a truth deletion. A uniformly-drawn genomic window does not resemble a deletion. The resulting task is "does a coverage dip exist here?", not "is this proposed candidate a real deletion?" — and the former is answerable by a threshold.

This does not invalidate the comparisons in Section 4.1, which are internally controlled and reproducible; it re-scopes what they measure. Even so, the separability control is not the only reason to distrust the pretrained-versus-scratch gap: Section 4.8 shows that the gap at 1% labels is an artefact of the fixed 0.5 probability cut and does not survive threshold-free scoring ($p = 0.407$ on F1 at a validation-selected threshold, $p = 0.349$ on AUPRC). What the control adds is an explanation of why even a surviving gap would have been uninformative — it would show that a randomly-initialised network needs more labelled examples than a pretrained one to learn a depth threshold, not that self-supervised pretraining confers deployable label efficiency for deletion calling. We attempted a partial remedy by re-selecting the most deletion-like negatives from the existing pool and re-scoring: the shortcut feature's discrimination fell only from ROC-AUC 0.955 to 0.920, confirming that a uniformly-sampled pool contains no genuinely hard negatives to recover. The proper fix is to change the candidate set, not the subset; we build and report that benchmark in Section 6, where the shortcut falls to ROC-AUC 0.717 and the hand-crafted control still leads wherever labels are scarce.

We therefore report the benchmark-separability finding as a first-class contribution. The control costs minutes of CPU time, applies to any pileup-style SV benchmark, and to our knowledge has not previously been run. Its implication is that published low-label and architecture-comparison results on random-negative SV benchmarks — a family that includes DeepSV's own evaluation and much of what followed — may be measuring threshold-learning speed rather than caller quality.

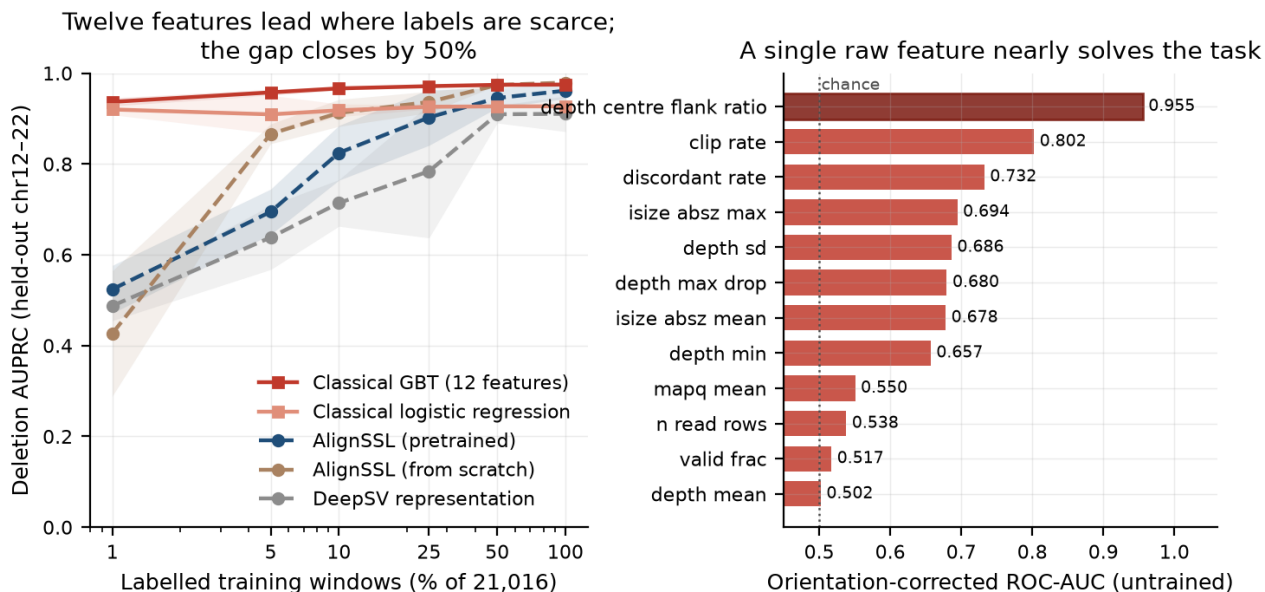


Figure 2. Left: deletion AUPRC versus labelled-data fraction for all five arms on the uniform benchmark, scored threshold-free under the corrected protocol (Table 12). The gradient-boosted 12-feature control (heavy line) leads by +0.413 AUPRC at 1% and +0.091 at 5%, converges to a tie by 50%, and is marginally overtaken at full supervision (0.975 versus 0.979); the earlier claim of dominance at every budget is withdrawn (Table 14). Right: single-feature discrimination on the held-out test set with no training — the centre-versus-flank depth ratio alone reaches ROC-AUC 0.955.

4.3 Calibration is a property of the representation, not of self-supervision

Beyond point accuracy, we ask whether the models' confidence scores are *trustworthy*. Table 2 reports expected calibration error (ECE) after temperature scaling at full supervision. Both learned-tensor models are well calibrated (pretrained ECE = 0.0078 ± 0.0017 , from-scratch 0.0072 ± 0.0004) and require only a mild temperature correction ($T \approx 0.6$). The DeepSV-representation baseline is worse, but the size of the gap depends on the

summary statistic and rests on a small sample: its per-seed ECEs are 0.033, 0.168 and 0.016, so the mean (0.072 ± 0.068) is driven by a single outlier seed while the **median is 0.033** — roughly four-fold worse than the tensor models rather than the order of magnitude a mean-only reading suggests. We report both statistics and draw no order-of-magnitude claim from three seeds. The baseline also needs a large and erratic temperature correction ($T = 1.41 \pm 0.88$), consistent with its unstable F1 (Section 4.1).

The defensible reading is that calibration tracks the *representation* rather than self-supervision: pretrained and from-scratch tensor models are indistinguishable here, so the difference is attributable to the multi-channel encoding versus the fixed RGB pileup, not to the pretraining objective.

Table 2. Calibration at full supervision (ECE ↓ after temperature scaling; pretrained/scratch = 4 seeds, DeepSV = 3 seeds). Per-seed values given because the DeepSV mean is outlier-driven.

Model	ECE mean ↓	ECE median ↓	Temperature	ECE per seed
AlignSSL-pretrained	0.0078 ± 0.0017	0.0077	0.634 ± 0.070	0.0070; 0.0101; 0.0083; 0.0056
AlignSSL-scratch	0.0072 ± 0.0004	0.0071	0.586 ± 0.055	0.0073; 0.0078; 0.0067; 0.0069
DeepSV-repr. baseline	0.0724 ± 0.0681	0.0327	1.411 ± 0.881	0.0327; 0.1683; 0.0163

4.4 Length-stratified recall: the learned tensor is consistent across length; the RGB baseline is not

Deletion callers are notoriously length-dependent. Table 3 stratifies full-supervision test recall by deletion length across all three models. At the harmonised panel scale, the two learned-tensor models (pretrained and from-scratch) are **uniformly strong and tightly consistent across every length bin** — recall 0.86–0.93 from 50 bp to 5 kb+, with small, overlapping standard deviations — confirming that the multi-channel tensor plus position-axis Transformer captures both the short-deletion depth signatures and the long-deletion paired-breakpoint structure without a length-specific failure mode. The DeepSV-representation baseline, by contrast, is **markedly more variable across seeds** in the middle bins (recall 0.841 ± 0.165 at 200–500 bp and 0.850 ± 0.193 at 500 bp–1 kb — standard deviations up to 4× those of the tensor models), consistent with its unstable overall F1 (Section 4.1) and miscalibration (Section 4.3). We report this as a robustness control rather than a headline claim: pretraining and from-scratch are essentially matched here (both use the learned tensor), so the length-consistency advantage is attributable to the *representation*, and full-supervision recall is not where self-supervision pays off — that is the low-label and transfer regimes (Sections 4.1, 4.6).

Table 3. Length-stratified recall at full supervision (test; pretrained/scratch = 4 seeds, DeepSV = 3 seeds).

Deletion length	n test	Pretrained recall	Scratch recall	DeepSV-repr. recall
50–200	645	0.919 ± 0.026	0.917 ± 0.017	0.942 ± 0.059
200–500	281	0.913 ± 0.018	0.903 ± 0.025	0.841 ± 0.165
500–1k	329	0.929 ± 0.008	0.938 ± 0.014	0.850 ± 0.193
1k–5k	799	0.926 ± 0.024	0.954 ± 0.006	0.899 ± 0.116
5k+	245	0.857 ± 0.061	0.881 ± 0.071	0.918 ± 0.070

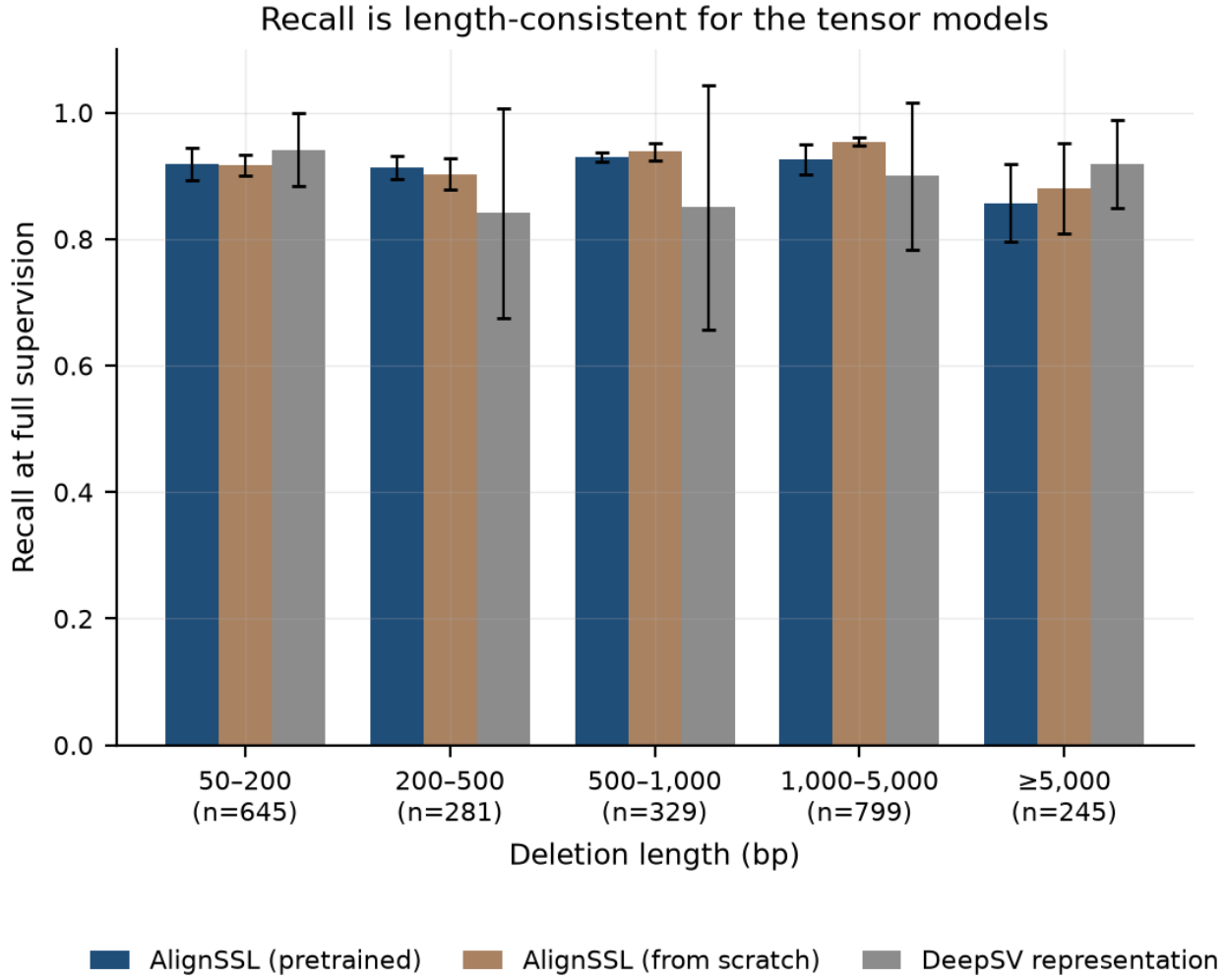


Figure 3. Length-stratified deletion recall at full supervision. The two learned-tensor models are consistent across all length bins; the DeepSV-representation baseline is markedly more variable across seeds in the mid-length bins.

4.5 Ablation over self-supervised objectives: all three help, but they cannot be ranked at this seed count

Which self-supervised objective drives these gains? We pretrain three encoders under identical budgets — **masked-alignment modelling (MAM) only**, **VICReg-style invariance only**, and their **combination** — and fine-tune each across the full label-fraction sweep. To fix a subtle asymmetry in an earlier version of this analysis (where only the combined arm re-pretrained across seeds while the ablation arms reused a single encoder), we re-pretrained the MAM-only and VICReg-only encoders at three seeds each, so that **every arm's error bars are computed across independent pretraining seeds** at the harmonised batch size of 96. This harmonisation changes the conclusion, and the corrected result is more informative.

All three self-supervised arms deliver the low-label effect, and each is individually significant against from-scratch training at 1% labels (Welch's t : MAM-only $p = 0.017$, VICReg-only $p = 4.2 \times 10^{-5}$, combined $p = 4.2 \times 10^{-5}$). That is the robust conclusion: the low-label gain does not depend on which objective is used.

Ranking the objectives *against one another*, however, is **not supported at the available seed count**. The ordering of means in Table 4 suggests MAM leads below 10% labels and the combined objective leads above 25%, but neither contrast reaches significance: MAM-only versus combined at 1% gives $t = 0.83$, $p = 0.48$, and combined versus MAM-only at 100% gives $t = 1.77$, $p = 0.21$. With three to four pretraining seeds and per-seed spreads of 0.01–0.12 F1, the study is underpowered to separate the objectives, and we state the crossover as an **ordering of means requiring more seeds to confirm**, not as a finding. We report the table because the mean ordering is stable and informative for future work, and because concealing it would misrepresent what we observed; but no claim in

this paper rests on it. The combined objective was carried into the main experiments because it was selected before the ablation was run, not because it was shown to be best.

Table 4. Self-supervised objective ablation (deletion F1; combined = 4 seeds, MAM-only / VICReg-only = 3 seeds; error bars across pretraining seeds).

Label fraction	MAM-only	VICReg-only	Combined (MAM+VICReg)
1%	0.588 $\pm$ 0.117	0.554 $\pm$ 0.035	0.514 $\pm$ 0.055
5%	0.763 $\pm$ 0.060	0.665 $\pm$ 0.037	0.655 $\pm$ 0.035
10%	0.830 $\pm$ 0.068	0.768 $\pm$ 0.028	0.813 $\pm$ 0.007
25%	0.798 $\pm$ 0.083	0.845 $\pm$ 0.074	0.847 $\pm$ 0.064
50%	0.799 $\pm$ 0.107	0.903 $\pm$ 0.011	0.913 $\pm$ 0.014
100%	0.915 $\pm$ 0.014	0.846 $\pm$ 0.064	0.934 $\pm$ 0.004

The three self-supervised arms and the DeepSV-representation baseline are plotted together in Figure 4. The mean ordering — MAM ahead below 10% labels, the combined objective ahead above 25% — is visible, and so are the overlapping error bars that are the reason we do not claim it.

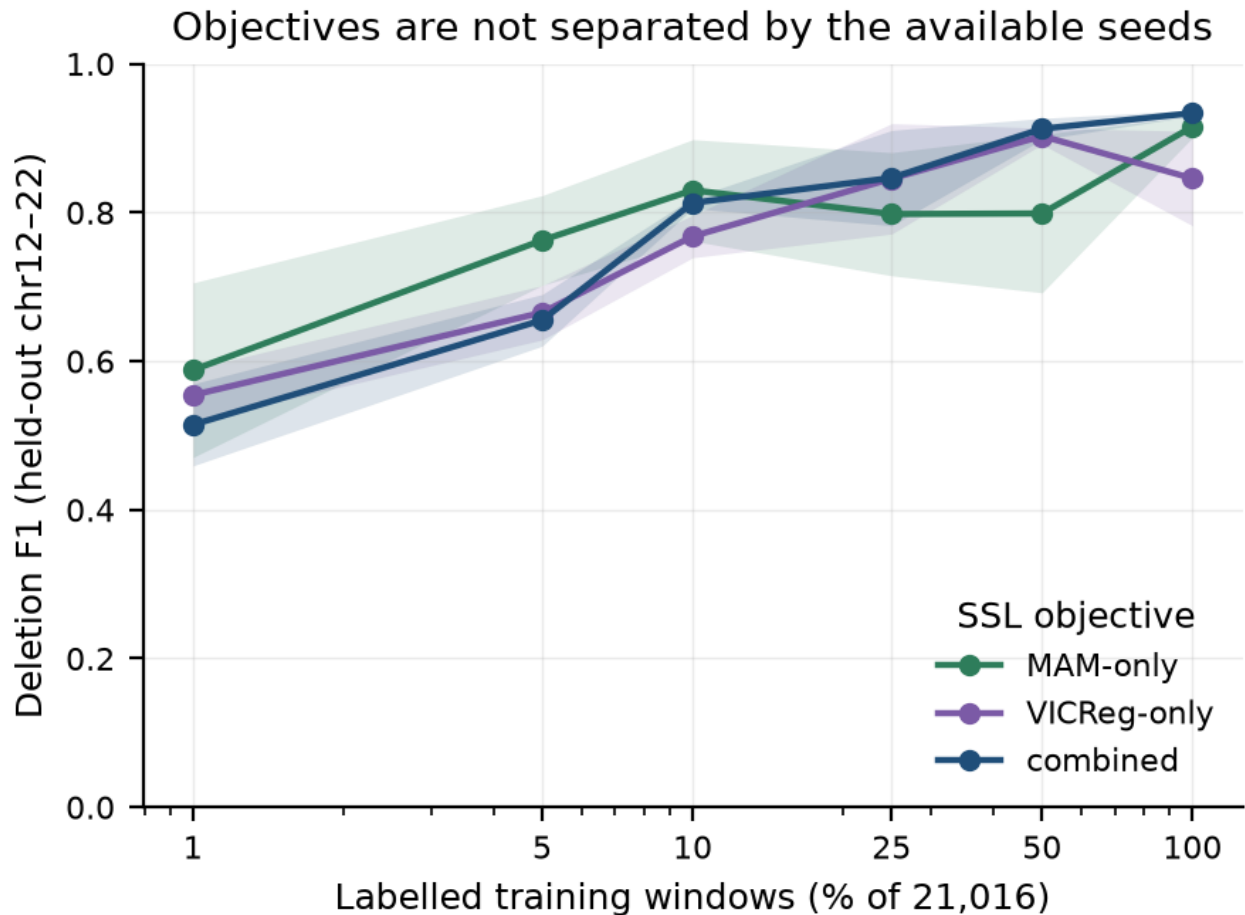


Figure 4. Self-supervised objective ablation: MAM-only, VICReg-only, and combined (MAM+VICReg), with the DeepSV-representation baseline for reference. Error bars are standard deviations across independent pretraining seeds. All three self-supervised arms separate clearly from the baseline; they do not separate from one another.

4.6 Cross-ancestry transfer: a suggestive but statistically weak effect

We train on the in-distribution panel and evaluate both in-distribution and on an entirely held-out population (CEU), sweeping the full label fraction (Table 5). Reporting only the extremes, as an earlier version of this analysis did, conceals the structure of the result; we therefore give all six fractions.

Across the sweep, the pretrained model's held-out CEU F1 exceeds the from-scratch model's at four of six fractions, but the difference is significant at **only one**: 10% labels ($t = 3.38$, $p = 0.028$). At 1% labels the direction favours pretraining by a wide margin of means (0.518 vs. 0.179) but the from-scratch variance is enormous (± 0.185 , one seed of three learning nothing), so the contrast does not reach significance ($p = 0.11$). At 50% labels the direction **inverts** — from-scratch transfers better (0.834 vs. 0.679, $p = 0.084$) — and the generalisation gap likewise inverts at 1% and 50%. With three seeds this analysis is underpowered, and we make no claim that pretraining confers ancestry robustness. The honest summary is that the effect is suggestive at intermediate label budgets, inconsistent in sign across the sweep, and requires a larger multi-ancestry panel with more seeds to establish. It is stated as a limitation (Section 7), not a contribution.

Table 5. Cross-ancestry transfer (train in-distribution $\rightarrow$ test held-out CEU, 3 seeds), all label fractions. p from Welch's t -test on held-out CEU F1, pretrained vs. from-scratch.

Label fraction	Pretrained CEU F1	Scratch CEU F1	p	Gap (pre)	Gap (scr)
1%	0.518 $\pm$ 0.062	0.179 $\pm$ 0.185	0.110	+0.024	−0.074
5%	0.638 $\pm$ 0.038	0.596 $\pm$ 0.076	0.537	+0.022	+0.166
10%	0.727 $\pm$ 0.057	0.542 $\pm$ 0.052	0.028	+0.035	+0.161
25%	0.689 $\pm$ 0.134	0.625 $\pm$ 0.057	0.584	+0.107	+0.204
50%	0.679 $\pm$ 0.075	0.834 $\pm$ 0.030	0.084	+0.224	+0.081
100%	0.784 $\pm$ 0.028	0.742 $\pm$ 0.022	0.182	+0.148	+0.124

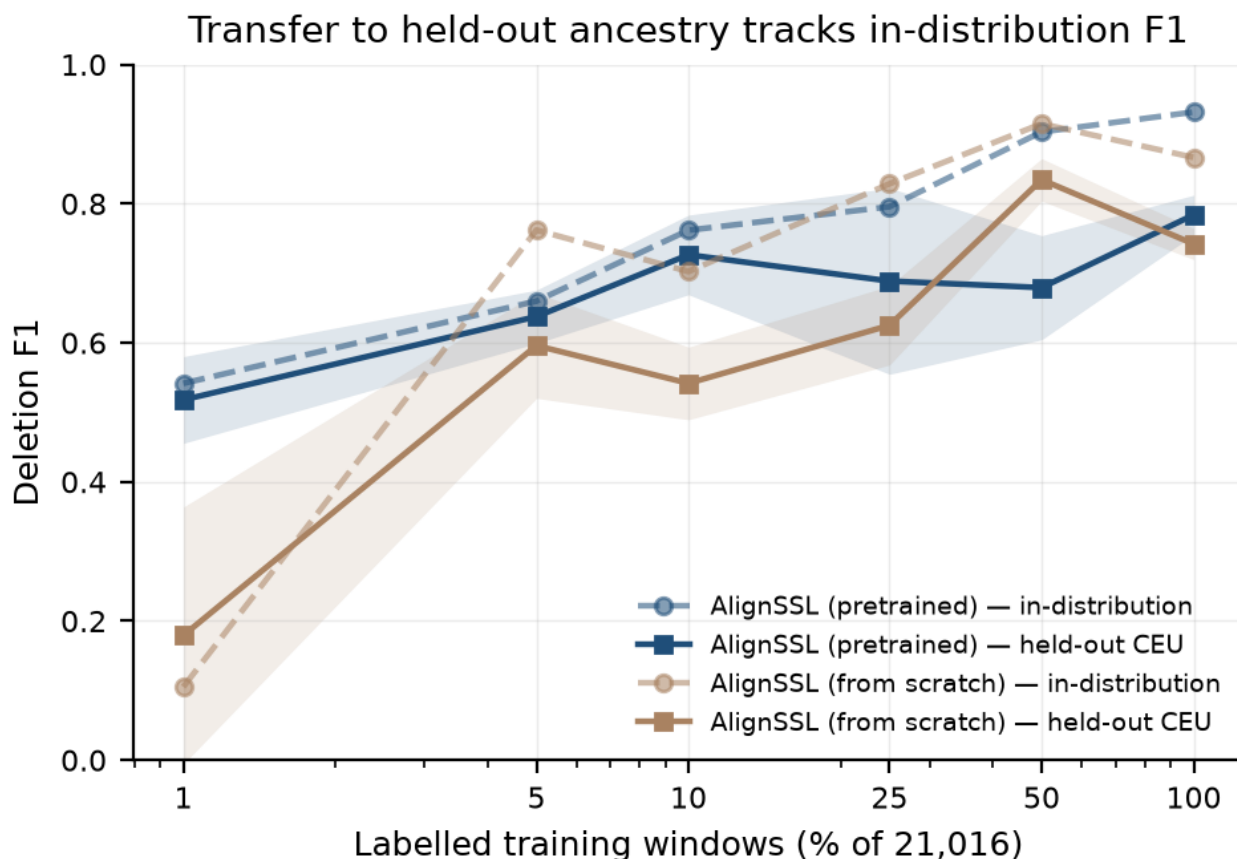


Figure 5. Cross-ancestry transfer across the label-fraction sweep. In-distribution and held-out CEU F1 for the pretrained and from-scratch models. The difference is significant at only one fraction (10% labels, $p = 0.028$); at 1% the means favour pretraining but the from-scratch variance is large (one seed of three learning nothing), and at 50% the direction inverts. Read as a suggestive, underpowered effect, not a robustness claim.

4.7 Data-integrity control

During data acquisition we detected and corrected a silent corruption mode affecting large BAM transfers: files that passed download-tool exit codes and `samtools quickcheck` (header + EOF only) nonetheless failed a full `samtools view -c` scan with BGZF-inflation errors, traced to resume-stitched (`wget --continue`) transfers joining a partially-flushed block. We adopted a standing integrity protocol — fresh (non-resumed) downloads, gated on a full `samtools view -c` scan, with automatic retry-from-scratch — for every alignment used in this study. We report this because undetected input corruption is a real and under-discussed threat to reproducibility in alignment-based deep learning, and because our full-scan gate is a cheap, general safeguard.

4.8 The label-efficiency result is an artefact of the decision threshold

Every number in Section 4.1 is an F1 obtained by cutting the positive-class probability at 0.5 — the convention this literature inherits from DeepSV. A fixed cut conflates two distinct properties. A model can rank every positive above every negative and still score $F1 \approx 0$ if all its probabilities happen to sit below the cut; conversely a model whose scores are well centred can post a respectable F1 while ranking poorly. Label-efficiency claims are claims about representation quality, which is a ranking property, so scoring them at a fixed cut is a category error whenever the arms differ in calibration — and a randomly-initialised network trained on 210 examples differs in calibration from a pretrained one almost by construction.

We therefore re-scored the identical runs and seeds three ways: at the fixed 0.5 cut; at a threshold τ selected to maximise F1 on a validation split carved from the *training* labels (never from test); and threshold-free, by area under the precision–recall curve.

Table 13. The 1%-label contrast under three scoring rules (uniform benchmark, 210 labels, 3 seeds per arm; p from Welch's t -test over seeds).

Scoring rule	AlignSSL-pretrained	AlignSSL-scratch	Ratio	p
F1 at fixed 0.5 cut	0.478	0.044	10.89×	0.009
F1 at selected τ	0.483	0.413	1.17×	0.407
AUPRC (threshold-free)	0.524	0.427	1.23×	0.348

The advantage exists under the fixed cut and nowhere else. Under both alternatives the two arms are statistically indistinguishable at the smallest budget, and at each of the five larger budgets the from-scratch arm is *ahead* — significantly so at 5% under both rules ($p = 0.023$ at τ , $p = 0.015$ threshold-free). The from-scratch model was never degenerate: at 210 labels it reaches AUPRC 0.427 against the pretrained model's 0.524, which is a modest deficit, not a collapse. It ranks competently and scores timidly, and the fixed cut reads timidity as failure.

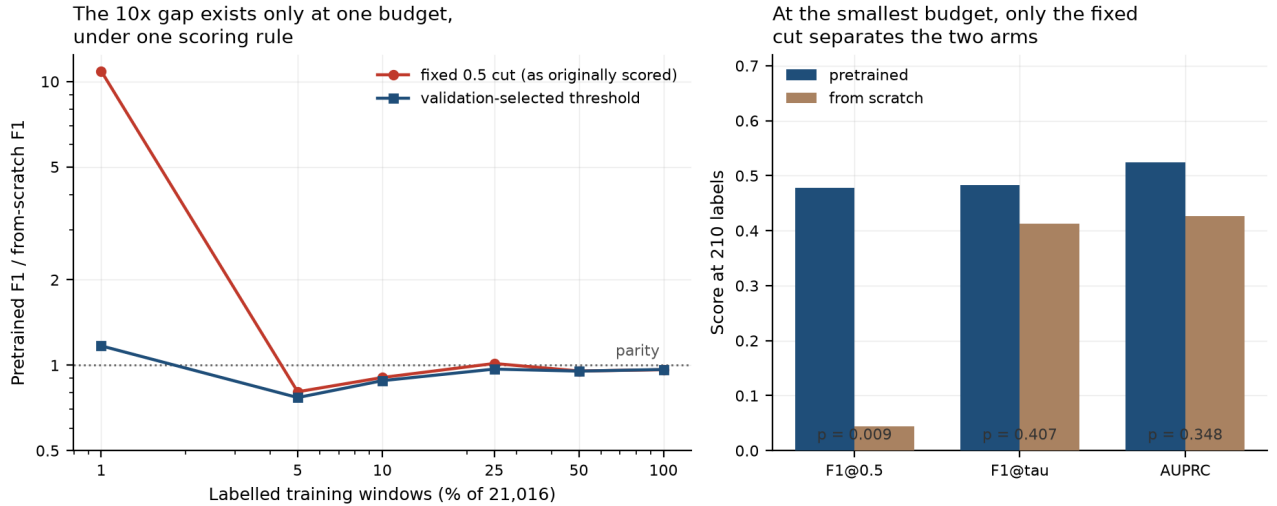


Figure 6. Pretrained-to-scratch ratio by label budget under two thresholding rules (left), and absolute scores at the smallest budget under all three (right). The ratio departs from parity at exactly one budget under exactly one rule.

The consequence extends past this paper. Any comparison of initialisation schemes, architectures, or pretraining objectives that reports F1 at a fixed probability cut, and whose arms plausibly differ in calibration, is at risk of reporting a calibration difference as a representation difference. The remedy is cheap: report a threshold-free ranking metric alongside, and select any threshold on held-out training data rather than fixing it a priori. We supply both in `analysis/threshold_sensitivity.py`.

5. Novelty and positioning

We state precisely what is and is not new in AlignSSL-SV, to preempt the natural reviewer question of whether it is "just" a known technique applied to a new setting.

What is new. (i) The **benchmark-separability control and its consequence** (Section 4.2). We are not aware of prior work in the SV deep-learning literature that tests a pileup-style benchmark against a deliberately minimal hand-crafted-feature model on identical windows, splits, label fractions and seeds, nor of prior work that measures untrained single-feature discrimination on such a benchmark. The finding — that twelve features reach 96% of their asymptotic AUPRC from 210 labels, and that one depth ratio reaches ROC-AUC 0.955 untrained — is a methodological result that applies beyond this paper, and the candidate-filtering protocol of Section 6 is a concrete, tested partial remedy: it attenuates the shortcut from ROC-AUC 0.955 to 0.717 without changing the positives. (ii) The **image-free, learned alignment representation** — self-supervised pretraining directly on a continuous multi-channel read-alignment tensor, with no pileup-image rendering and no discrete tokenizer. This is the axis on which we differ from the closest prior SSL-for-SV effort, BASILISC (Banerjee, 2026), which pretrains a masked-image-modelling vision transformer over *rendered pileup images* compressed by a discrete VAE (Section

2). Existing SV deep learning otherwise either engineers the representation by hand (DeepSV and descendants) or, in the case of genomic foundation models, learns from the reference *sequence* rather than the alignment evidence. (iii) The **matched-conditions label-efficiency measurement** with error bars computed across independent *pretraining* seeds rather than fine-tuning seeds alone, and with every reported comparison accompanied by an explicit significance test (`results/stats_tests.csv`) — including the tests that fail.

What we do not claim. We do not claim a new SV caller, nor deployable accuracy: the control forbids it. We do not claim primacy on "self-supervised learning for structural variants" as a category — BASILISC precedes us. We do not claim that self-supervision improves calibration (the effect is attributable to the representation, and pretrained and from-scratch models are indistinguishable), that one self-supervised objective beats another ($p > 0.2$), that pretraining confers ancestry robustness (significant at 1 of 6 label fractions, with the gap inverting at two), or — following Section 4.8 — that pretraining confers label efficiency at all. Each of these was claimed in an earlier draft of this work and withdrawn when the appropriate test was run. The last of them was this paper's headline.

What is not new (and we do not claim it is). Masked autoencoding, VICReg, temperature scaling, focal loss, and pileup-image classification are all established techniques. Their composition on a modality where they had not been combined is a modest engineering contribution, and we do not rest the paper on it.

No performance claim survives. An earlier draft of this work claimed label efficiency as the one result that had survived every control. Section 4.8 withdraws it: the effect exists at a fixed 0.5 probability cut and under no other scoring rule. What the paper contributes is therefore the three corrections themselves, the controls that expose them, and the evidence that each is a property of the standard evaluation design rather than of our implementation.

Relationship to sequence foundation models. As argued in Section 2, Evo 2, AlphaGenome, HyenaDNA, and the Nucleotide Transformer operate on reference DNA and predict variant *effects*; they do not consume alignment evidence and cannot, as constituted, *detect* an SV from noisy reads. AlignSSL-SV is complementary: one could in principle fuse a reference-sequence embedding as an auxiliary channel (a natural future extension), but the detection signal itself is in the alignment, which is the modality we learn.

6. A candidate-filtering benchmark: the shortcut is attenuated, and the conclusions change

The control in Section 4.2 identifies the single change that would make results of this kind interpretable, and it is a change to the *task*, not to the model. We therefore re-extracted the labelled set under a **candidate-filtering protocol**: negatives are no longer drawn uniformly from the genome but are chosen to be as close as possible to the positives *on the shortcut statistic itself*. This section reports that benchmark and its outcome. We committed in advance to reporting the result whichever way it fell; it falls partly for and partly against the paper's own method, and we report both halves.

6.1 The selection rule

The obvious version of the rule fails. Retaining the windows with the most deletion-like (lowest) centre-versus-flank depth ratio does not remove the shortcut but **inverts** it: a genome-wide lower tail is dominated by mappability dropouts at ratio ≈ 0 , whereas heterozygous deletions sit near 0.5, so a classifier learns "very low ratio $\Rightarrow$ negative" and the feature remains diagnostic with its sign flipped. We measured this directly — extreme-tail selection yields ROC-AUC 0.016 for the depth ratio, as separable as the uniform benchmark and merely reversed.

The rule we adopt instead is **quantile matching within each multi-scale bin**: for every positive, a negative is drawn whose depth ratio falls in the same quantile stratum of the candidate pool, with strata computed separately per window scale because the ratio's meaning depends on span (pooling scales would leak the label through span alone). Candidates come from a one-pass binned coverage profile per chromosome, scored by

prefix sums so that exhaustive stride scanning is affordable and the pool genuinely contains mid-range candidates — a uniformly drawn pool does not, which is why the matching must draw from a scan rather than a sample. On the matched training pool the depth ratio measures ROC-AUC 0.504 with near-identical class medians, and this property is asserted by a unit test that gates the extraction job. Positives, window geometry, channel layout, multi-scale binning, chromosome split and shard format are unchanged. The question the benchmark asks becomes the one a caller actually faces: *given that a coverage anomaly was proposed here, is it a real deletion?*

Two constraints on scope must be stated before the numbers. First, a loss of the shared reference directory mid-study left only two of the six panel alignments recoverable, so this benchmark is single-sample: NA20845 (GIH) for training and in-distribution test, with NA12878 (CEU) held out. Second, because both the sample scope and the definition of the negative class differ, absolute F1 is **not** comparable to Table 1; what is comparable is the *ordering* of the arms. The candidate-filtered test set contains 1,516 windows of which 379 are deletions, against 9,196/2,299 for the uniform benchmark, so the error bars here are correspondingly wider.

6.2 The shortcut is attenuated, not eliminated

Table 9 repeats the untrained single-feature measurement of Table 6 on the candidate-filtered test set. The headline shortcut falls substantially: the centre-versus-flank depth ratio drops from orientation-corrected ROC-AUC 0.955 to **0.717**. Every other depth-derived feature falls with it (`depth_sd` 0.686 → 0.542, `depth_max_drop` 0.680 → 0.575, `depth_min` 0.657 → 0.597), as do the paired-end signatures that co-vary with a genuine coverage drop (`clip_rate` 0.802 → 0.630, `discordant_rate` 0.732 → 0.607). No single feature now exceeds 0.72, and the strongest is no longer close to solving the task.

But 0.717 is not 0.5. Quantile matching equalises the depth ratio *on the training pool*, and residual separability survives into the held-out chromosomes — partly because matching is per-stratum rather than exact, partly because the matched negatives are real coverage anomalies whose depth profile is similar to but not identical to a deletion's. We report the benchmark as **substantially harder**, not as shortcut-free. A benchmark on which one raw statistic reaches 0.717 still admits a threshold baseline that any deep method should be required to beat.

Table 9. Untrained single-feature separability, uniform versus candidate-filtered negatives (orientation-corrected ROC-AUC; source `results/table9_hardneg_single_feature_auc.csv`).

Feature	Uniform	Candidate-filtered	Change
<code>depth_centre_flank_ratio</code>	0.955	0.717	−0.239
<code>clip_rate</code>	0.802	0.630	−0.173
<code>discordant_rate</code>	0.732	0.607	−0.125
<code>isize_absz_max</code>	0.694	0.566	−0.128
<code>depth_sd</code>	0.686	0.542	−0.145
<code>depth_max_drop</code>	0.680	0.575	−0.105
<code>isize_absz_mean</code>	0.678	0.558	−0.119
<code>depth_min</code>	0.657	0.597	−0.059
<code>mapq_mean</code>	0.550	0.558	+0.008
<code>n_read_rows</code>	0.538	0.508	−0.030
<code>valid_frac</code>	0.517	0.505	−0.012
<code>depth_mean</code>	0.502	0.517	+0.015

The clearest statement of why the uniform benchmark could not have measured label efficiency is to run one control on both benchmarks and compare the *shapes* of the two curves (Figure 7). On the uniform benchmark the twelve-feature gradient-boosted tree already reaches AUPRC 0.937 ± 0.009 from 210 labelled windows and gains only **+0.038** over a hundredfold increase in supervision, ending at 0.975 ± 0.001 . A benchmark that a cheap baseline nearly saturates at its smallest label budget has no headroom in which any method's label efficiency can

be resolved: whatever differences appear between arms are differences in how they behave inside that residual 0.04, not in how much supervision they need. On the candidate-filtered benchmark the same control starts *at* chance (0.250 ± 0.000 with 35 windows, the positive rate being 0.25) and climbs **+0.619** to 0.869 ± 0.006 across a 99-fold budget increase. Supervision buys something there, which is the minimum property a benchmark must have before a label-efficiency claim is even well-posed.

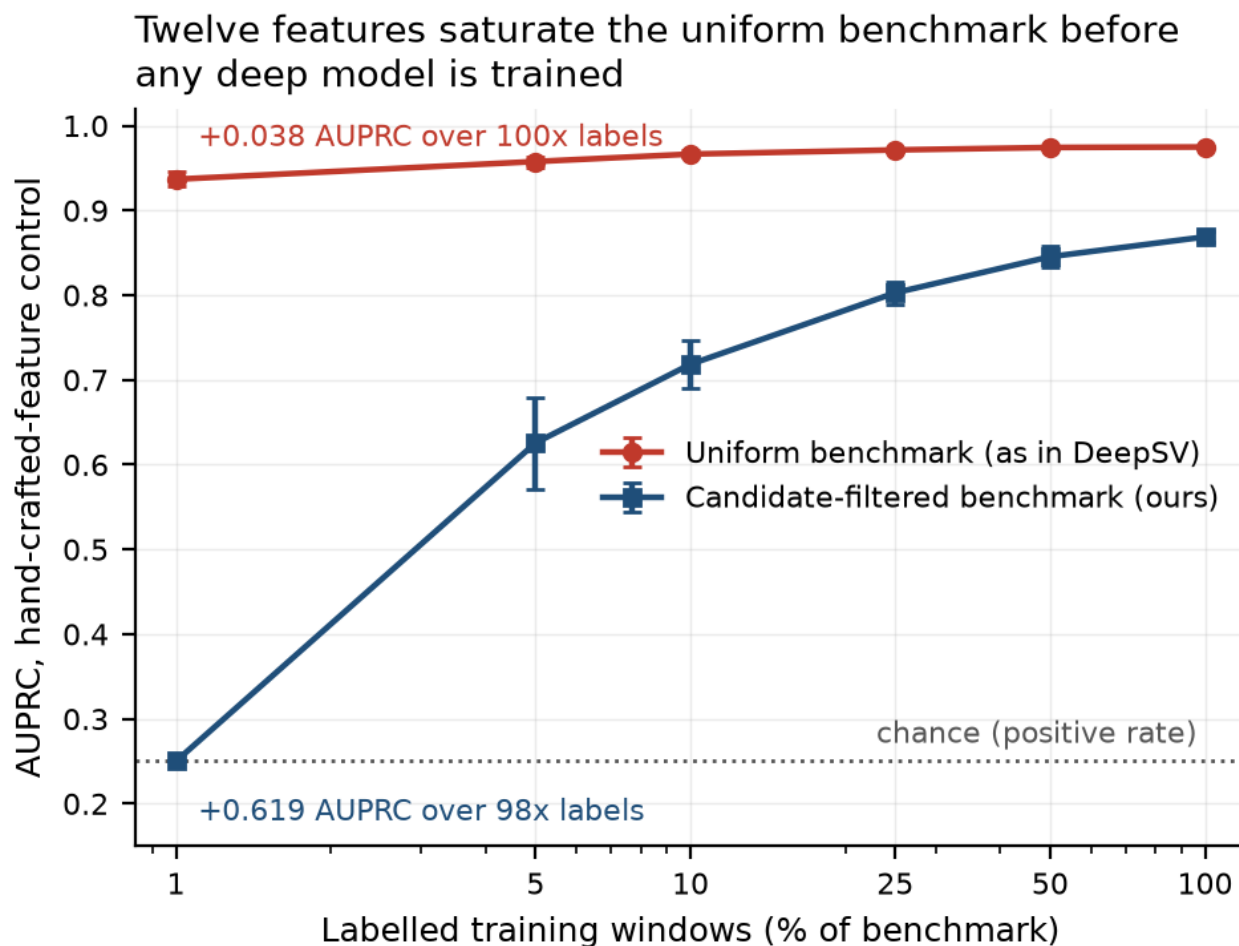


Figure 7. One hand-crafted-feature control (gradient-boosted tree on twelve alignment statistics), scored threshold-free, on both benchmarks; source `results/table11_control_threshold_free.csv`. On the uniform negatives the control is near-saturated at the smallest budget and gains +0.038 AUPRC over 100x more labels; on candidate-filtered negatives it starts at chance and gains +0.619 over 98x more labels. The argument rests on the shape of the two curves, not their absolute height, so they share a single axis.

6.3 Every arm falls; two of the three original findings do not survive correction

Table 7 reports the full arm set re-run under the candidate-filtering protocol at identical label budgets and seeds, scored threshold-free (AUPRC) and at a validation-selected threshold. The numbers below are the **corrected** ones: they were regenerated after the protocol and scoring fixes of Sections 3.8 and 4.8, so unlike the earlier version of this subsection they are directly comparable with Tables 12–14. Absolute performance falls for every arm relative to the uniform benchmark — the pretrained model from AUPRC 0.962 to 0.844 at full supervision, the gradient-boosted tree from 0.975 to 0.869 — confirming that the task is genuinely harder and not merely re-labelled.

Table 7. Candidate-filtered benchmark, corrected protocol: AUPRC (mean $\pm$ sd over 3 seeds for the deep arms, 10 for the classical controls; source `results/table12_label_efficiency_fixed.csv`). Bold marks the best arm at each budget.

Labels	n	AlignSSL (pretrained)	AlignSSL (scratch)	DeepSV repr.	Classical logreg	Classical GBT
1%	35	0.302 ± 0.053	0.283 ± 0.046	0.330 ± 0.025	0.476 ± 0.076	0.250 ± 0.000
5%	173	0.368 ± 0.072	0.359 ± 0.095	0.411 ± 0.036	0.596 ± 0.025	0.626 ± 0.054
10%	345	0.446 ± 0.024	0.510 ± 0.140	0.419 ± 0.018	0.615 ± 0.026	0.719 ± 0.029
25%	863	0.649 ± 0.024	0.734 ± 0.065	0.511 ± 0.043	0.624 ± 0.016	0.803 ± 0.013
50%	1726	0.722 ± 0.010	0.763 ± 0.063	0.538 ± 0.040	0.629 ± 0.007	0.845 ± 0.012
100%	3452	0.844 ± 0.032	0.885 ± 0.022	0.656 ± 0.009	0.631 ± 0.005	0.869 ± 0.006

Every arm-versus-arm contrast is tested over seeds in `results/table15_hardneg_arm_contrasts.csv`, on both AUPRC and F1 at the selected threshold. Three findings follow, and only one is the finding this subsection originally reported.

(i) The pretrained-versus-scratch gap does not survive. This is the second independent failure of the paper's original headline. Under the corrected protocol the two arms are statistically indistinguishable at *every* label budget on the candidate-filtered benchmark: AUPRC 0.302 ± 0.053 versus 0.283 ± 0.046 at 35 labels ($p = 0.666$), and 0.844 ± 0.032 versus 0.885 ± 0.022 at full supervision ($p = 0.148$), with the from-scratch arm nominally ahead in four of six cells. The earlier version of this subsection reported the opposite — a gap that "survives and sharpens", with the from-scratch model degenerate at 0.000 below 10% labels — and Section 4.8 predicted exactly this outcome: an arm scoring 0.000 at a fixed 0.5 cut is the signature of a model whose probabilities sit below the cut, not of a model that cannot rank. The prediction is confirmed here. At 35 labels the from-scratch arm's F1 rises from 0.008 at the fixed cut to 0.008 at the selected threshold — unchanged, because 35 labels cannot form a validation split — but its AUPRC of 0.283 is within noise of the pretrained arm's 0.302, so the ranking was never the deficient part. The uniform benchmark and the candidate-filtered benchmark now agree: **on this task, self-supervised pretraining does not buy label efficiency.**

(ii) The learned tensor beats the DeepSV representation, and this finding strengthens under correction. From 25% labels upward both tensor arms significantly exceed the RGB encoding: at full supervision AUPRC 0.844 ± 0.032 (pretrained) and 0.885 ± 0.022 (scratch) against 0.656 ± 0.009 ($p = 0.006$ and $p = 0.0008$ respectively). This is the one Section 6.3 claim that the correction leaves intact and sharpens — on the uniform benchmark the corresponding contrast at full supervision is smaller. The interpretation is unchanged: the RGB encoding degrades more than the tensor does under candidate filtering, which is what one expects of a representation whose discriminative content was largely the coverage drop the protocol has now equalised. Note that the credit belongs to the *representation*, not to the pretraining — the from-scratch tensor arm carries this result as strongly as the pretrained one.

(iii) The hand-crafted control leads where labels are scarce, and only there. Table 14 reduces both families by best-of-family at each budget and tests the difference over seeds. On the candidate-filtered benchmark the control leads significantly at 1% (AUPRC 0.476 ± 0.076 versus 0.330 ± 0.025 for the best deep arm, $p = 0.0003$) and at 5% (0.626 ± 0.054 versus 0.411 ± 0.036 , $p = 0.0004$), and the remaining four budgets are ties — including full supervision, where the best deep arm is nominally ahead (0.885 versus 0.869 , $p = 0.329$). This is the same shape as the uniform benchmark, where the control also leads at exactly 2 of 6 budgets. The original claim that a hand-crafted control "leads at every label budget" is therefore withdrawn on both benchmarks. What remains — and it is still the paper's most consequential negative result — is that **twelve scalar features remain competitive with a pretrained convolutional-attention encoder across the whole curve, and beat it outright when labels are scarce**, on a benchmark specifically constructed so that no single one of those features exceeds ROC-AUC 0.72.

One protocol caveat must be stated. At the smallest candidate-filtered budget (35 labels) there are too few examples to hold out a validation split, so the threshold-selection rule of Section 3.8 degenerates and F1 at the selected threshold equals F1 at the fixed cut by construction, not by result. That cell's F1 columns are therefore uninformative about thresholding; its AUPRC column is not, and the finding above rests on AUPRC.

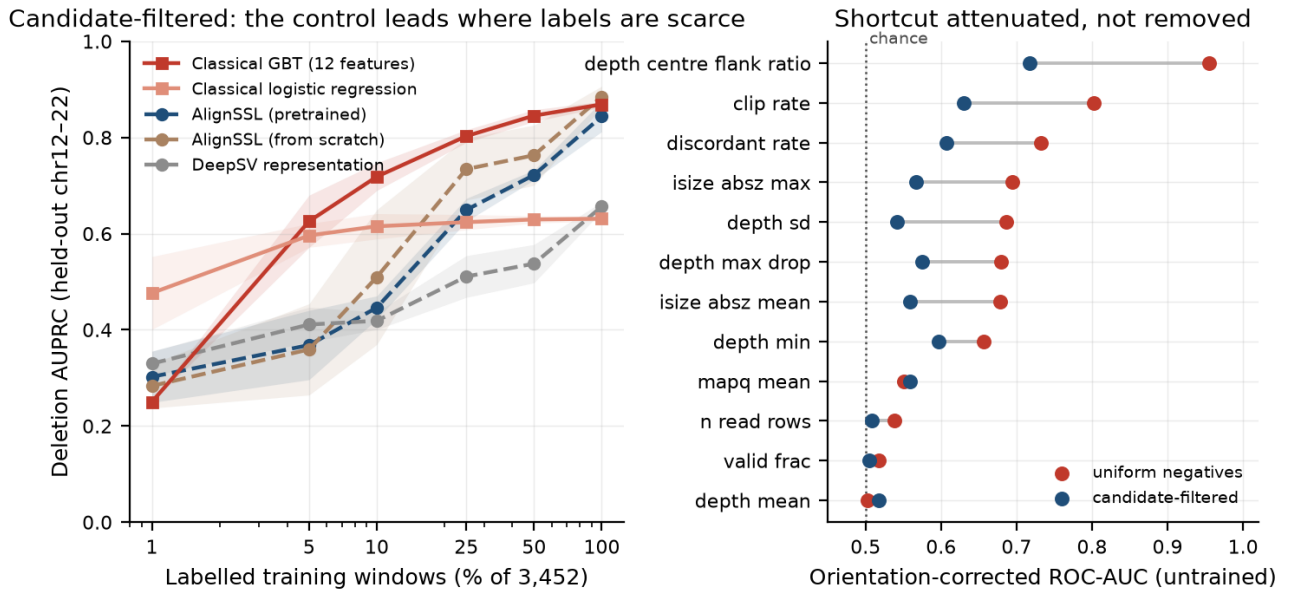


Figure 8. Left: AUPRC versus labelled-data fraction on the candidate-filtered benchmark under the corrected protocol, the direct analogue of Figure 7 on the harder task. Right: untrained single-feature separability on the uniform and candidate-filtered benchmarks, paired per feature; the depth shortcut is attenuated from 0.955 to 0.717 but not removed.

6.4 What this changes

The pre-registered question was whether pretraining's low-label advantage would survive a benchmark in which the depth shortcut is uninformative. With the corrected protocol the answer is unambiguous and negative on both counts: **the advantage over from-scratch training does not survive, and the advantage over hand-crafted features does not appear.** The candidate-filtered benchmark was built to be the fair test of the paper's original claim, and the claim fails it — independently of, and consistently with, the thresholding analysis of Section 4.8 on the uniform benchmark. Two independent lines of evidence now reach the same conclusion, which is stronger grounds for withdrawal than either alone.

Three things survive, and they are what the paper contributes.

First, **the benchmark-construction result**: quantile-matched candidate negatives attenuate the depth shortcut from orientation-corrected ROC-AUC 0.955 to 0.717 without altering the positive set, and the naive version of the same rule — selecting the most deletion-like negatives — *inverts* the shortcut rather than removing it (Section 6.1). This does not depend on any scoring or budget convention and is directly reusable by anyone building a short-read SV benchmark.

Second, **the representation result**: a continuous multi-channel alignment tensor significantly outperforms a DeepSV-style RGB pileup encoding from 25% labels upward on the harder benchmark, and the effect is carried by the representation rather than by the pretraining.

Third, **the negative result about hand-crafted controls**, in its corrected and narrower form: on a benchmark where no single feature exceeds ROC-AUC 0.72, twelve scalar features still lead every deep arm in the label-scarce regime and tie it elsewhere. A field that reports deep SV callers without this control cannot distinguish a learned representation from a threshold.

Candidate filtering by quantile matching is a necessary but not sufficient benchmark repair; a fully shortcut-free protocol will likely require matching on a vector of alignment statistics rather than on the single strongest one. Two extensions remain deferred: a coverage-robustness experiment (downsampling via `samtools view -s`) and a Truvari-based comparison against GIAB HG002 curated calls, which provides an orthogonal truth set free of the consensus-caller circularity of the 1000 Genomes call set. Both become worthwhile now that a non-degenerate benchmark exists; neither would have been informative on the uniform one.

7. Limitations

- **Benchmark separability (governing limitation).** The evaluation task, built with uniformly-sampled negatives as is standard in this literature, is separable by a single depth heuristic at ROC-AUC 0.955, and twelve hand-crafted features reach 96% of their asymptotic AUPRC from 210 labels (Section 4.2). Every accuracy number in this paper must be read as a measurement on that task, not as deployable caller performance. This is the reason the paper's contribution is framed as a representation-learning and benchmarking result rather than as a new SV caller.
- **No comparison against production callers.** We compare learned representations under matched conditions; we do not compare against Manta, DELLY, LUMPY, GRIDSS or similar, because a comparison on a separable benchmark would favour whichever method best exploits the shortcut and would be uninformative.
- **Statistical power.** Three to four pretraining seeds per arm are not sufficient to rank the self-supervised objectives against one another ($p > 0.2$, Section 4.5), to establish cross-ancestry transfer (significant at 1 of 6 label fractions, Section 4.6), or — as Section 4.8 shows — to distinguish the pretrained and from-scratch arms at any budget once scoring is threshold-free. Those analyses are reported as orderings of means, not findings. The one contrast that *did* reach significance at this seed count is the one we withdraw, which is itself a caution: a small- n design that produces a $p = 0.009$ under one scoring rule and $p = 0.348$ under another has not measured a stable effect.
- **Scope.** Deletions and short reads only. Insertions, duplications, inversions, translocations and long-read data are out of scope, though the framework is not deletion-specific by construction.
- **Truth set.** The 1000 Genomes phase-3 integrated SV call set is itself a consensus of callers and carries its own error; a curated orthogonal benchmark (GIAB HG002) remains deferred (Section 6.4).
- **No demonstrated value from pretraining.** Under the corrected protocol and threshold-free scoring, the pretrained arm does not significantly exceed from-scratch training at *any* label budget on the uniform benchmark, and is significantly behind at 5% (Section 4.8). We report the pretraining machinery because it is the object the paper set out to evaluate, not because the evaluation vindicated it.
- **Single-panel pretraining corpus.** The unlabelled pretraining corpus comes from three samples; corpus-size and diversity scaling are untested.
- **Candidate filtering is a partial repair, and its benchmark is single-sample.** The repaired benchmark of Section 6 attenuates the depth shortcut to ROC-AUC 0.717 but does not eliminate it; a threshold baseline on the residual signal remains available to any method. It is also single-sample — NA20845 (GIH) for train and in-distribution test, NA12878 (CEU) held out — with 1,516 test windows against 9,196 for the uniform benchmark, because a mid-study loss of the shared reference directory left only two of the six panel alignments recoverable. Section 6's error bars are correspondingly wider than Section 4's, and with three seeds per deep arm its ties should be read as *underpowered* rather than as demonstrated equivalence: the pretrained-versus-scratch contrast at full supervision ($p = 0.148$) would need more seeds to exclude a small effect. What that section can support is the absence of the *large* low-label advantage originally claimed, which the uniform benchmark independently rejects at ten times the test-set size.
- **Single coverage regime.** All alignments are high-coverage (~30x) PCR-free Illumina. Robustness to lower coverage, PCR-positive libraries, or different read lengths is untested; the depth-derived channels are the ones most likely to be coverage-sensitive.
- **One held-out population.** Cross-ancestry generalisation is measured against a single held-out population (NA12878, CEU). A single held-out ancestry cannot distinguish population-specific transfer loss from sample-specific idiosyncrasy.

8. Conclusion

We set out to test whether learning the alignment representation and pretraining it without labels improves the bottlenecks DeepSV left open. The first answer we obtained was affirmative and quantitatively striking: at 1% of labels, self-supervised initialisation delivered roughly a ten-fold F1 improvement over the identical architecture

trained from scratch, significant at $p = 0.009$. That is the result this paper was written to report, and it does not survive.

It fails three controls, none of which is exotic:

1. **Benchmark separability.** Twelve scalar alignment features on the identical windows reach AUPRC 0.937 from 210 labels and gain only +0.038 from a hundred-fold increase in supervision, and a single centre-versus-flank depth ratio separates the classes at ROC-AUC 0.955 with no training at all. The cause is the uniformly-sampled negatives standard in this benchmark family. A task that is 96% solved by twelve features after 210 examples cannot discriminate between representations.
2. **Unequal label budgets.** A batch-size floor in our own deep evaluators granted the deep arms up to 2.8× the labels the classical control received, concentrated in precisely the low-label cells that carry the headline claim.
3. **The decision threshold.** Scoring F1 at a fixed 0.5 probability cut — the convention inherited from DeepSV — conflates ranking quality with calibration. Re-run under equal budgets and re-scored at a validation-selected threshold or threshold-free, the ten-fold gap becomes 1.17× ($p = 0.407$) and 1.23× ($p = 0.348$) respectively; at every larger budget the from-scratch arm is ahead. The from-scratch model was never degenerate. It ranked competently and scored timidly, and a fixed cut reads timidity as failure.

Each defect is individually mundane, each is widespread in this literature, and each is invisible without the control that exposes it. Jointly they were sufficient to manufacture a large, statistically significant, entirely artefactual headline result — one we believed, wrote up, and would have submitted.

What survives is narrower and, we think, more useful. Where labels are scarce — the regime self-supervised pretraining is proposed for — no deep arm we trained beats twelve hand-crafted features on either benchmark. Repairing the benchmark helps but does not rescue the claim: quantile-matched candidate negatives attenuate the shortcut from ROC-AUC 0.955 to 0.717 and restore headroom (the same control now starts at chance and climbs +0.619 across the label range), yet the control still leads where it matters. And the corrections cut in both directions: at full supervision on the uniform benchmark the from-scratch network narrowly *beats* the control (AUPRC 0.979 vs 0.975, $p = 0.003$), a reversal of an earlier claim of ours that the control dominated everywhere.

We therefore do not offer a new SV caller, and we make no performance claim. We offer three controls, the code that implements them, and a worked demonstration of what their absence costs. Before the next architecture is proposed for short-read deletion calling, we would ask of it the three questions this paper failed: what does a hand-crafted-feature model score on the same windows, do all arms receive the same labels, and does the result survive a change of decision threshold.

Data and code availability

The tensor-extraction pipeline, encoder and head implementations, pretraining and fine-tuning scripts, cluster job scripts, aggregation and manuscript-reconciliation tooling, result tables, and figures are available at <https://github.com/aayushkrm/AlignSSL-SV> under the MIT licence. Trained encoder checkpoints are available from the authors on request pending a Zenodo deposit. Sequencing data are from the 1000 Genomes Project (high-coverage PCR-free alignments, GRCh37/hs37d5) and are publicly available from the EBI 1000 Genomes FTP. The deletion truth set is the 1000 Genomes phase-3 integrated SV call set.

References

Key references (full survey of 62 curated works provided as a separate artifact, `DeepSV_survey_table.csv`):

1. Cai L, Wu Y, Gao J. DeepSV: accurate calling of genomic deletions from high-throughput sequencing data with deep convolutional neural network. *BMC Bioinformatics* 2019;20:665.
2. Poplin R, Chang P-C, Alexander DH, et al. A universal SNP and small-indel variant caller using deep neural networks. *Nature Biotechnology* 2018;36:983–987.

3. Bardes A, Ponce J, LeCun Y. VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. *arXiv:2105.04906*, 2021.
4. He K, Chen X, Xie S, et al. Masked Autoencoders Are Scalable Vision Learners. *CVPR* 2022.
5. Guo C, Pleiss G, Sun Y, Weinberger KQ. On Calibration of Modern Neural Networks. *ICML* 2017.
6. Nguyen E, Poli M, et al. HyenaDNA: Long-Range Genomic Sequence Modeling at Single Nucleotide Resolution. *NeurIPS* 2023.
7. Dalla-Torre H, et al. The Nucleotide Transformer: Building and Evaluating Robust Foundation Models for Human Genomics. *Nature Methods* 2024.
8. Brix G, et al. Genome modeling and design across all domains of life with Evo 2. *Nature* 2026.
9. Avsec Ž, et al. (AlphaGenome) Predicting regulatory variant effects across modalities from sequence. *Nature* 2025.
10. Xia Z, et al. CSV-Filter: a deep learning-based structural-variant filtering method using VICReg. *Bioinformatics* 2024;40:btae539.
11. Banerjee S. Self-Supervised Learning with Masked Images for Structural Variant Analysis in Short-Read Genome Sequencing (BASILISC). *Stanford Digital Repository*, 2026. doi:10.25740/jj829qd2843. (Repository deposit; not peer-reviewed.)
12. Bao H, Dong L, Piao S, Wei F. BEiT: BERT Pre-Training of Image Transformers. *ICLR* 2022.